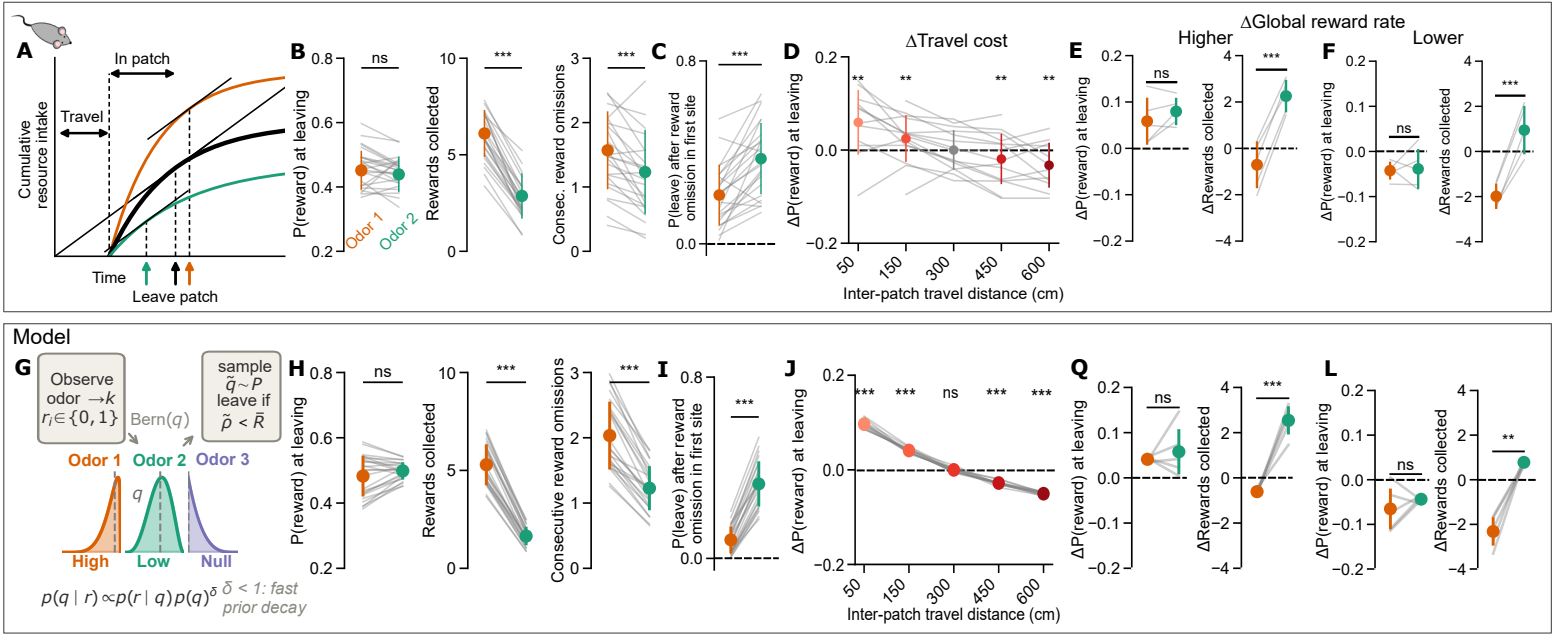

A
In patch
Travel
Cumulative resource intake
Time
Leave patch
B
P(reward) at leaving
Odor 1
Odor 2
ns
Rewards collected
***
Consec. reward omissions
C
P(leave) after reward omission in first site
D
ΔTravel cost
ΔP(reward) at leaving
Inter-patch travel distance (cm)
E
Higher
ΔGlobal reward rate
ΔP(reward) at leaving
ΔRewards collected
F
Lower
Model
G
Observe odor $\rightarrow k$
$r_i \in \{0,1\}$
Bern(q)
sample $\tilde{q} \sim P$
leave if $\tilde{p} < \bar{R}$
Odor 1
Odor 2
Odor 3
q
High
Low
Null
$p(q \mid r) \propto p(r \mid q)p(q)^{\delta}$
$\delta < 1$: fast prior decay
H
I
J
Q
L
Consecutive reward omissions